Solving quadratic equations

1 Solve the following equations:

a $3y - 15y^2 = 0$

b $x^2 + 6x - 55 = 0$

c $4x^2 - x - 10 = 0$

d $8 - 7m - m^2 = -2m^2 + m + 2$

e $x - \dfrac{45}{x} = 4$

f $\dfrac{10}{x} - 3x = -13$

2 State whether each of the following are solutions of the equation $2x^2 - 7 = 0$:

a $x = \pm\sqrt{\left(\dfrac{7}{2}\right)}$

b $x = \pm\dfrac{\sqrt{14}}{7}$

c $x = \pm\dfrac{\sqrt{14}}{2}$

d $x = \pm\sqrt{\left(\dfrac{2}{7}\right)}$

3 Consider the equation $x\,(x + 9) = -20$.

a Solve it by the method of factorisation.

b Check your solution by solving it using the quadratic formula.

Applications

4 A circle has equation $(x + 2)^2 + (y - 4)^2 = 41$. Find the x-intercepts of the circle.

5 On Earth, the equation $d = 4.9t^2$ is used to find the distance (in meters) an object has fallen through the air after t seconds.

a 8 seconds after a skydiver jumps out of the plane, what distance has she fallen?

b Solve for t, the time it will take her to fall 78.4 metres.

6 On the moon, the equation $d = 0.8t^2$ is used to approximate the distance an object has fallen after t seconds. (Assuming no air resistance or buoyancy). On Earth, the equation is $d = 4.9t^2$.

a A rock is thrown from a height of 60 metres on Earth. Solve for t, the time it will take to hit the ground to the nearest second.

b If a rock is thrown from a height of 60 metres on the moon, solve for the time, t, it will take to reach the ground to the nearest second.

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7 The Gateway Arch in St. Louis, Missouri, is $630\ \text{ft}$ tall.
We can model the behaviour of objects falling from the arch using Galileo's formula for falling objects: $d = 16t^2$, where d is distance fallen in feet and t is time in seconds since the object was dropped.
How long would it take an object dropped from the arch to fall to the ground? Give your answer to the nearest second.

8 The kinetic energy E of a moving object is given by $E = \frac{1}{2}mv^2$, where m is its mass in kilograms and v is its speed in metres/second.

a If a vehicle travelling at $18\ \text{m/s}$ has kinetic energy $E = 307\,800$, what is the mass of the car?

b If a vehicle is travelling at $18\ \text{m/s}$, what is its speed in kilometres/hour to one decimal place?

c If a vehicle weighing 1600 kilograms has kinetic energy $E = 204\,800$, at what speed is it moving?

9 The sum of the series $1 + 2 + 3 + \ldots + n$ is given by $S = \frac{n\,(n+1)}{2}$.

a Find the sum of the first 32 positive integers.

b Find the number, n, of positive integers required for a sum of 15.

10 In a room of n people, if everyone shakes hands with everyone else, the total number of handshakes is given by $\frac{n\,(n-1)}{2}$. If 780 handshakes are made, how many people are there in the room?

11 The product of 2 positive consecutive numbers is 56. Let the smaller number be x.

a Solve for x.

b Find the larger number.

12 A number increased by 3 is multiplied by the same number decreased by 2, and the result is 50. Using x to represent the number, write an equation that represents the situation and solve it for x.

13 Mae throws a stick vertically upwards. After t seconds, its height h metres above the ground is given by the formula $h = 25t - 5t^2$.

a At what time(s) will the stick be 30 metres above the ground?

b How long does the stick take to hit the ground?

c Can the stick ever reach a height of 36 metres?

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14 A t-shirt is fired straight up from a t-shirt cannon at ground level. After t seconds, its height above the ground is h metres, where $h = -16t^2 + 48t$.

a For what values of t is the shirt 11 metres above the ground?

b Hence, calculate how long the t-shirt was at least 11 metres above the ground.

c For what values of t is the shirt 35 metres above the ground?

d Hence, calculate how long the t-shirt was at least 35 metres above the ground.

15 The perimeter of a rectangle is $28\ \text{cm}$. Let the length be $x\ \text{cm}$ and the width be $y\ \text{cm}$.

a Find an equation for y in terms of x.

b Given that the area of the rectangle is $48\ \text{cm}^2$, find all of the possible values of x.

c Find all of the possible values for y.

16 The area of a rectangle is $160\ \text{m}^2$. The length of the rectangle is $6\ \text{m}$ longer than its width. Let w be the width of the rectangle in metres.

a Find the width of the rectangle, w.

b Find the rectangle's length.

17 At time t seconds, the distance, s, travelled by an object moving in a straight line is given by

$$s = ut + \frac{1}{2}at^2$$

where u is its starting speed and a is its acceleration.
When $u = 16$ and $a = 8$, find how long it would take for the object to travel 128 metres.

18 The base of a triangle is 3 metres more than twice its height. The area of the triangle is 115 square metres. Let x be the height of the triangle.

a Find the height by solving for x.

b Find the length of the base.

19 The product of 2 consecutive, positive odd numbers is 399.
Let the smaller number be $2x + 1$.

a Solve for x.

b Find the first odd number.

c Find the other odd number.

20 The square of a number is 70 less than 17 times the number. Let x be the number.
What is the largest possible number that satisfies this condition?

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21 The sum of the areas of two circles is $79.56\pi \text{ m}^2$. One of the circles has a radius 1.1 times the length of the other. Let r be the length of the shorter radius.

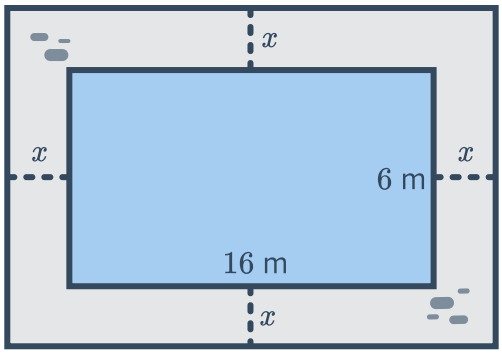
a Find the length of the shorter radius.

b Find the length of the longer radius.

22 A rectangular swimming pool is 16 m long and 6 m wide. It is surrounded by a pebble path of uniform width $x \text{ m}$. The area of the path is 104 m^2 .

a Find an expression for the area of the path in terms of x. Express your answer in simplest form.

b Hence, form an equation and solve for x, the width of the path.



23 Georgia can break a code in 5 hours less time than her younger sister Tricia. If they work together, the code will take them 6 hours to break.

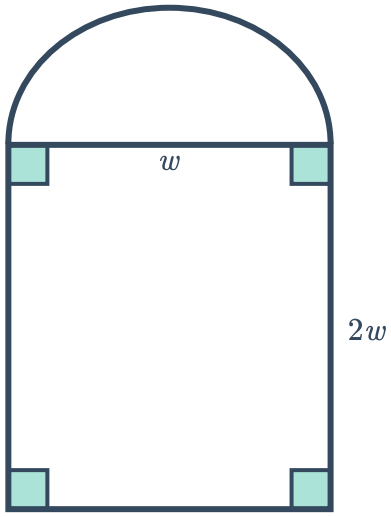
a Find the number of hours, m, that it would take Georgia to break the code.

b Hence, find how long it would take Tricia to crack the code.

24 The diameter of a semicircle shares its edge with the width of a rectangle whose length is twice its width. Their combined area is 13 cm^2 .

a Find the width of the rectangle correct to two decimal places. Let w be the width.

b Hence, find the perimeter of the entire shape correct to one decimal place.



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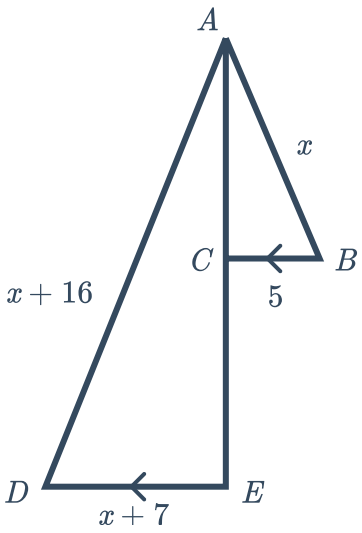
25 The sum of two integers is 70.

- **a** If one of the integers is 29, what is the other integer?
- **b** Let one of the integers be x. Write an expression for the other integer.
- **c** Hence, write an expression for the product of the two integers. Give your answer in expanded form.
- **d** When $x = 35$, what is the product of the two integers?
- **e** When $x = 35$, does this result in the largest possible product?

26 We want to investigate which points have a y-coordinate 7 and are 5 units from $(2, 3)$.

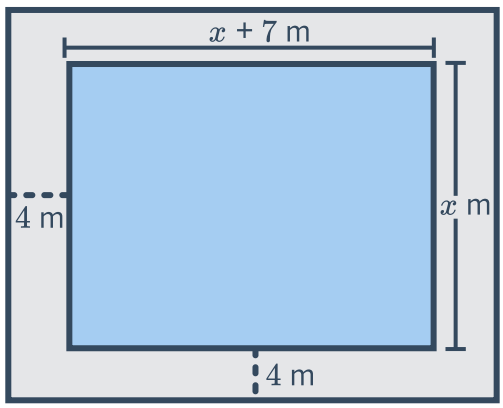
- **a** Find the x-coordinate(s) of the points.
- **b** Write the coordinates of the points.

27 In the diagram, $\triangle ABC$ is similar to $\triangle ADE$, with BC parallel to ED.
Find the value of x.



28 The rectangular pool in the image has a length of $x + 7$ metres and a width of x metres. It is surrounded by a path 4 metres wide.

- **a** Find an expression for the area of the pool in terms x.
- **b** Find an expression for the total area of the pool and surrounding path, in terms of x.
- **c** Suppose the total area of the pool and the surrounding path is $488\ \text{m}^2$ more than the area of the pool alone. Find the value of x.
- **d** Hence, state the width and length of the pool.



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Further quadratic equations

29 Solve the following equation for x by substituting in $m = 4^x$.

$$4^{2x} - 65 \times 4^x + 64 = 0$$

30 Consider a random fraction $\frac{x}{y}$ which has the property that the sum of itself and its reciprocal is $\frac{25}{12}$.

a Express the sum of the fraction and its reciprocal as a single fraction in terms of x and y.

b Luke decided to set up a pair of equations by considering the numerator and denominator of the fraction from part (a). From the numerator, he formed the equation:

$$x^2 + y^2 = 25$$

Form an equation involving x and y from the denominator.

c Hence, solve the equations for all possible values of x.

d Find y for each of these values of x.

e Given that the fraction $\frac{x}{y}$ lies between 0 and 1, find $\frac{x}{y}$.

31 Consider the continued fraction:

$$x = \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ldots}}}$$

a Which of the following equations is correct according to the above fraction:

A $x = \frac{1}{x}$ **B** $1 = \frac{x}{x+1}$ **C** $x = \frac{1}{1+x}$

b Solve the equation from part (a) to find the exact value of the continued fraction x.

32 The following continued surd has an exact value.

$$x = \sqrt{8 + \sqrt{8 + \sqrt{8 + \sqrt{8 + \ldots}}}}$$

Find the exact value of x in simplest surd form.

33 Solve the following equations.

a $\left(x^2 + 2x + 2\right)^{x^2+5x+4} = 1$

b $\left(x^2 + 3x - 9\right)^{x^2+x-6} = 1$

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